

Degree Project in the Field of Technology Media Technology and the Main Field of Study Computer Science and Engineering

Second cycle, 30 credits

Human-Centered AI Assistance in Design Tools

Designing a Task-Aware Co-Pilot to Support UX Designers' Exploration and Decision-Making

MARTA CASANDRA LENDÍNEZ IBÁÑEZ

Human-Centered AI Assistance in Design Tools

Designing a Task-Aware Co-Pilot to Support UX Designers' Exploration and Decision-Making

MARTA CASANDRA LENDÍNEZ IBÁÑEZ

Master's Programme, Interactive Media Technology, 120 credits
Date: August 29, 2026

Supervisor: Ran Zhou
Examiner: Jarmo Laaksolahti
School of Electrical Engineering and Computer Science
Swedish title: Människocentrerad AI-assistans i designverktyg
Swedish subtitle: Att designa en uppgiftsmedveten AI-co-pilot för att stödja UX-designers utforskning och beslutsfattande

Human-Centered AI Assistance in Design Tools

Designing a Task-Aware Co-Pilot to Support UX Designers' Exploration and Decision-Making

CASANDRA IBÁÑEZ

Recent advances in Artificial Intelligence (AI), particularly Large Language Models (LLMs), have significantly expanded the capabilities of digital tools. While AI-powered assistants are widely used in software development, their integration into User Experience (UX) design workflows remains limited. Existing AI-driven design tools often prioritize automation over collaboration, which can reduce designers' control, transparency, and sense of ownership.

This thesis investigates how AI can support UX designers without replacing their agency. Instead of treating AI as an automated system, the work explores AI as a collaborative co-pilot that assists designers during exploration and decision-making. The study combines literature review, qualitative research, and design prototyping to examine current practices and challenges in AI-assisted design.

Insights from expert interviews and prior research informed the development of a conceptual framework for human-centered, task-aware AI assistance. Based on this framework, a prototype AI plugin was developed to demonstrate how AI can support designers while maintaining transparency and human control. The thesis contributes empirical insights, a framework for task-aware AI assistance, and a prototype illustrating AI as a collaborative design partner.

SAMMANFATTNING

De senaste framstegen inom artificiell intelligens (AI), särskilt stora språkmodeller (LLM), har avsevärt utökat möjligheterna i digitala verktyg. Även om AI-baserade assistenter används i stor utsträckning inom mjukvaruutveckling är integrationen av AI i UX-designprocesser fortfarande begränsad. Befintliga AI-drivna designverktyg prioriterar ofta automatisering framför samarbete, vilket kan minska designers kontroll, transparens och känsla av ägarskap.

Denna uppsats undersöker hur AI kan stödja UX-designers utan att ersätta deras handlingsutrymme. I stället för att betrakta AI som ett automatiserat system utforskar arbetet AI som en samarbetande "co-pilot" som stödjer designers i utforskande och beslutsfattande. Studien kombinerar litteraturstudier, kvalitativ forskning och designprototyping för att analysera nuvarande arbetssätt och utmaningar inom AI-assisterad design.

Insikter från expertintervjuer och tidigare forskning användes för att utveckla ett konceptuellt ramverk för människocentrerad och uppgiftsmedveten AI-assistans. Baserat på detta ramverk utvecklades en prototyp av ett AI-plugin som demonstrerar hur AI kan stödja designers samtidigt som transparens och mänsklig kontroll bibehålls. Uppsatsen bidrar med empiriska insikter, ett ramverk för uppgiftsmedveten AI-assistans och en prototyp som illustrerar AI som en samarbetspartner i design.

Author's address: Casandra Ibáñez, mcli2@kth.se, KTH Royal Institute of Technology, School of Electrical Engineering and Computer Science, Stockholm, Sweden, SE 100 44.

Document date: August 29, 2026

Swedish title: Människocentrerad AI-assistans i designverktyg

Swedish subtitle: Att designa en uppgiftsmedveten AI-co-pilot för att stödja UX-designers utforskning och beslutsfattande

© 2022 Copyright held by the author.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**; *Interaction design*; Collaborative interaction.

Keywords: Human-Centered Artificial Intelligence, UX Design Tools, AI-Assisted Design, Large Language Models, Human–AI Collaboration

Nyckelord: Människocentrerad artificiell intelligens, UX-designverktyg, AI-assisterad design, Stora språkmodeller, Människa–AI-samarbete

1 INTRODUCTION

Artificial Intelligence (AI) is increasingly integrated into professional tools and workflows. Recent advances in Large Language Models (LLMs) have enabled AI systems to support tasks such as programming, writing, and information retrieval [3, 17]. In software development, AI-powered assistants are now embedded directly into development environments, assisting with code generation, debugging, and documentation [6, 22]. These developments illustrate a broader shift toward AI-augmented work, where digital tools participate not only in execution but also in decision-making processes [3**?**].

Despite this rapid progress, the integration of AI into User Experience (UX) design workflows remains limited. Existing AI-driven design tools often focus on automated generation of user interfaces or visual assets [7, 9, 23]. While these systems can accelerate production, they frequently prioritize automation over collaboration, reducing designers' sense of control, transparency, and ownership over the design process [14, 18, 24]. Designers also report challenges such as inaccurate outputs, hallucinated interface elements, and difficulty translating design intent into generated results [4, 5, 15].

This gap raises important questions about how AI should be integrated into human-centered design practices. Design work relies on iterative exploration, reflection, and critical evaluation—processes that risk being diminished when AI systems automate large parts of the workflow [1, 19]. When designers receive fully generated outputs with limited insight into how they were produced, they may become passive recipients rather than active participants in exploration and decision-making [16, 25]. This reduces opportunities for reflective thinking and weakens designer agency [2, 8].

These challenges motivate the central research question of this thesis:

What design principles enable task-aware AI assistance to support UX designers' exploration and decision-making while preserving transparency and human agency?

The purpose of this thesis is to encourage a more critical and human-centered approach to integrating AI into UX design tools. Through a conceptual framework and a prototype AI-assisted tool, the work aims to illustrate how AI can function as a collaborative "co-pilot" that supports designers' thinking rather than replacing it. The insights from this research provide guidance for developing AI systems that enhance, rather than diminish, human agency in design workflows.

2 RELATED WORK

This section reviews prior research on AI-assisted design, focusing on three areas relevant to this thesis: the use of generative AI in early-stage UX workflows, human–AI collaboration, and emerging AI-powered design tools. Together, these domains provide the foundation for understanding current opportunities and limitations in AI-supported UX design.

2.1 AI in UX Design Processes

Generative AI has introduced new possibilities for supporting early-stage UX design workflows, particularly during ideation and prototyping. Prior work shows that generative UI tools can help designers overcome the blank-page problem by rapidly producing initial layouts, wireframes, or stylistic variations that stimulate exploration across multiple directions [12, 21]. These systems enable designers to externalize partially formed ideas quickly, supporting divergent thinking before manual refinement. However, the outputs produced by such tools are rarely production-ready and typically require substantial human intervention [12, 21].

Beyond interface generation, generative AI has been explored as a catalyst for early ideation. Studies highlight how AI-generated suggestions can introduce alternative perspectives, expand the design space, and encourage designers to consider directions they might not have explored independently [1]. Yet despite these benefits, most systems focus on producing interface-level solutions rather than supporting upstream activities such as problem framing, contextual understanding, or design reasoning. Designers often rely on their own expertise or external tools to articulate the design problem before engaging with generative systems [21].

Recent work further suggests that large language models do not reliably improve problem reframing. Controlled experiments comparing AI-assisted and manual reframing found no significant improvement in novelty or usefulness when AI support was used, indicating that contextual reasoning and domain understanding remain challenging for current models [20]. Together, these findings suggest that while generative AI can accelerate early exploration, it offers limited support for deeper cognitive processes central to UX design.

2.2 Human–AI Collaboration in Design

A parallel line of research examines AI as a collaborator rather than an autonomous generator. Designers frequently describe AI tools as creative assistants that support idea exploration while leaving interpretation, evaluation, and decision-making firmly in human hands [11]. This collaborative framing aligns with broader HCI guidelines emphasizing transparency, predictability, and user control as prerequisites for trust in AI-powered systems [5]. When AI behavior is opaque or unpredictable, designers' confidence in the system can erode quickly.

Studies on collaborative creativity highlight persistent challenges in trust calibration. Designers often misjudge AI capabilities—either overestimating or underestimating what the system can do—which can negatively affect collaborative performance and shape how designers engage with AI-generated suggestions [2]. These findings reinforce the need for AI systems that empower designers rather than automate creative work, supporting human agency throughout the process.

2.3 AI Design Tools and Systems

Several research efforts propose systems that integrate AI into design workflows in more structured ways. Akin demonstrates how deep learning can generate UI wireframes from pattern libraries, producing structured layouts that serve as starting points for further refinement [9]. More recent work explores AI-powered copilots that assist designers across multiple stages of the design process. DesignManager, for example, coordinates multiple AI capabilities within a unified workflow, enabling designers to interact with different forms of AI assistance as their tasks evolve [26].

Other work focuses on helping designers conceptualize AI as a design material. Toolkits such as AI Opportunity Cards provide structured prompts and guidelines to help designers identify meaningful opportunities for AI integration while avoiding unnecessary or inappropriate uses [12]. These systems aim to scaffold designers' reasoning about when and how AI should be incorporated into interactive experiences.

2.4 Current Challenges and Research Gaps

2.4.1 Challenges and Limitations of Generative AI in Design. Despite growing interest in AI-assisted design tools, researchers have identified several limitations and risks. Generative models often recombine existing design patterns rather than producing genuinely novel ideas, raising concerns about homogenization and reduced diversity in interface design [16]. Hallucinations and inconsistencies are also common: AI-generated outputs may appear plausible but contain errors, misinterpretations, or unintended biases, affecting both the generated content and the resulting user experience [18].

Concerns have also been raised about skill erosion, particularly for junior designers. Over-reliance on AI tools may reduce opportunities to develop foundational design skills and critical thinking abilities [5]. Additionally, studies show that AI assistance can influence how people approach creative tasks, sometimes reducing depth of engagement, idea elaboration, or reflective thinking [20].

2.4.2 Interaction Challenges in AI Design Tools. Beyond conceptual limitations, current AI design tools present significant interaction challenges. Many systems rely heavily on text-based prompting, which designers often find misaligned with visual and iterative design practices. Translating visual intent into precise textual descriptions can be difficult, making prompt writing a barrier to effective use [21]. Designers consistently express a preference for more interactive, visual, and manipulable forms of AI interaction.

Accessibility presents another challenge. Evaluations of generative web builders show that even when accessibility requirements are explicitly included in prompts, resulting designs still contain numerous violations [13]. Similarly, research on prompt-based accessibility improvements finds that increasing prompt complexity alone does not reliably improve accessibility outcomes [10]. These findings suggest that effective AI-assisted design tools require stronger constraints, embedded domain knowledge, and interactive guidance mechanisms rather than relying solely on natural language prompts.

3 RESEARCH METHODS

This thesis adopts a multi-method research strategy that integrates literature review, qualitative inquiry, conceptual framework construction, prototype development, and empirical evaluation. Each method contributes a distinct layer of evidence: the literature review establishes the theoretical foundation, the formative study provides empirical grounding, the framework synthesises these insights into a structured model, the prototype operationalises the model, and the evaluation examines how designers interact with the system in practice.

3.1 Literature Review

The research began with a structured review of prior work on generative AI in UX design, human–AI collaboration, AI-powered design tools, and interaction challenges in AI-assisted creativity. Searches were conducted through academic databases such as the ACM Digital Library, IEEE Xplore, and Google Scholar, using keywords related to AI-assisted design, UX workflows, and reflective design practices.

Papers were selected based on their relevance to early-stage design processes, transparency, designer agency, and interaction models. The review identified recurring challenges such as over-automation, hallucinations, misaligned prompting, and reduced engagement. These insights shaped the initial problem framing and highlighted the need for a task-aware, human-centered approach to AI assistance.

3.2 Formative Expert Study

To complement the theoretical foundation with real-world perspectives, a formative study was conducted with five experts in UX design, software engineering, and human–AI interaction research. Participants were recruited through purposive sampling to ensure familiarity with AI-supported workflows. Semi-structured interviews were used to explore how experts currently use AI tools, where they find value, and where they encounter limitations. The interviews also examined perceptions of trust, control, and appropriate use. All sessions were conducted remotely and documented through detailed notes. Thematic analysis was applied to identify cross-cutting patterns across participants, revealing themes related to early-stage usefulness, risks of cognitive offloading, challenges in trust calibration, uncertainty about appropriate use cases, and frustrations related to reliability. These themes provided the empirical grounding for the framework.

3.3 Framework Construction

The framework was developed through an abductive process that combined insights from the literature review, the formative expert study, and a small design experiment conducted as part of this thesis. In addition to analysing prior work and interview findings, I carried out a structured comparison of designing a feature both with and without AI assistance. The exercise involved creating the same loyalty-flow feature under two conditions—one using only traditional design methods and one incorporating AI support—while maintaining a detailed log of decisions, artefacts, prompts, and moments of friction.

Comparing these two workflows provided first-hand insight into how AI alters designers' reasoning, attention, and engagement. The No-AI condition highlighted the manual structuring and iterative reasoning typical of early-stage design, whereas the AI condition revealed tendencies toward cognitive offloading, shifts in direction prompted by AI suggestions, and challenges in translating visual intent into textual prompts.

By triangulating insights from literature, expert perspectives, and this practice-based comparison, I identified recurring patterns in how designers work with and without AI support. These patterns informed the structure of the framework, which organises AI-assisted design into six workflow stages, Intent, Context, Options, Critique, Improve, and Trace.

3.4 Prototype Development

To operationalise the framework, a functional web-based prototype was developed using a research-through-design approach. The design process involved iterative refinement of interface sketches, interaction flows, and system behaviours. The frontend was implemented using React, TypeScript, Vite, TailwindCSS, and Framer Motion, and was responsible for orchestrating the six-stage workflow, rendering AI-generated options and critiques, animating reasoning sequences, and supporting interactive manipulation of design elements. The backend was implemented in Node.js and Express, acting as a stateless intermediary between the frontend and a Groq-hosted LLaMA-based model. It received structured requests containing intent and context, forwarded them to the model, and returned structured responses

containing options, reasoning, critiques, and summaries. This architecture ensured modularity and allowed the model layer to be replaced without altering the interaction design.

3.5 Evaluation

The prototype was evaluated through two complementary usability studies conducted at different stages of fidelity. The first evaluation focused on a low-fidelity version of the system and examined whether the six-stage workflow was understandable, whether the interaction model aligned with designers' expectations, and whether the overall structure supported reflective reasoning. The second evaluation tested the high-fidelity functional prototype and investigated how designers interacted with the implemented features, how they interpreted AI-generated options and critiques, and how effectively the system supported exploration and decision-making in practice.

Across both studies, participants were recruited through convenience sampling and had prior experience using AI tools in design contexts. Sessions were conducted remotely and lasted between 45 and 60 minutes. In each session, participants completed a design task involving the improvement of an onboarding flow for a fictional application. They were encouraged to think aloud while working, allowing observation of how they navigated the workflow, interpreted system outputs, and responded to the AI's reasoning. Following the task, participants took part in a short interview exploring their perceptions of usefulness, transparency, control, and overall experience.

4 FORMATIVE STUDY

The goal of the study was to understand how designers and developers currently use AI tools, how they reason about trust and control, and what barriers emerge when integrating AI into creative or analytical workflows. Insights from this study informed the design goals and requirements for the framework presented in the next chapter.

4.1 Participants and Procedure

I recruited five experts with professional experience in UX design, software engineering, or human–AI interaction research. Participants were selected through purposive sampling on LinkedIn to ensure familiarity with AI-supported workflows and exposure to real-world design or development practices. The sample included both industry practitioners and academic researchers, allowing me to capture perspectives from applied and theoretical domains.

Each interview followed a semi-structured format. This approach maintained a consistent set of themes while allowing questions to adapt to each participant's expertise. Topics included current uses of AI in design or development workflows, areas where AI tools provide value, situations where AI is less helpful, perceptions of control, risks associated with AI-assisted work, and expectations for future AI design tools. All interviews were conducted remotely and documented through detailed written notes.

4.2 Analysis

I analyzed the interview notes using qualitative thematic analysis. After each session, I extracted key observations and compared them across participants to identify converging themes. Rather than focusing on individual accounts, the analysis aimed to capture broader patterns in how experts evaluate AI tools, negotiate trust and control, and adapt their workflows in response to AI-generated content.

4.3 Findings

The analysis revealed several recurring themes that characterize how experts experience AI-assisted design and development workflows.

4.3.1 AI Supports Early-Stage Exploration. Participants consistently described AI as most useful during preparatory and exploratory stages of work. AI tools were used to draft interview questions, organize information, generate initial ideas, or support early research. In these contexts, AI acted as a productivity amplifier—helping structure work, reduce repetitive effort, and accelerate early divergence. However, participants emphasized that AI's usefulness decreases as tasks become more complex or require deeper contextual understanding. Detailed design work, privacy-sensitive tasks, or decisions requiring nuanced domain knowledge were seen as areas where human judgment remains essential.

4.3.2 Over-Automation Reduces Engagement. A major concern across interviews was the risk of over-automation. Participants noted that when AI generates solutions quickly, users may become less engaged in reasoning through the problem. One participant described this as "cognitive offloading," where the system performs the thinking and the user becomes a passive evaluator. Others noted that large AI-generated outputs—such as long code blocks or multiple design suggestions—can make it difficult to maintain a clear mental model of how the solution works. These observations suggest that the challenge is not only incorrect output but interaction models that encourage passive acceptance rather than critical engagement.

4.3.3 Trust and Miscalibrated Confidence. Trust emerged as a central theme. Participants noted that conversational AI systems often present answers in confident, polished language, which can lead users to overestimate the reliability of outputs. This miscalibration creates two problematic behaviors: over-reliance on AI suggestions or excessive verification effort. Several participants described strategies for managing this tension, such as distrusting outputs by default or manually checking the reasoning behind them.

4.3.4 Unclear Boundaries of Appropriate Use. Participants frequently expressed uncertainty about when AI should or should not be used. While AI was seen as helpful for some tasks, it was considered unsuitable for others—particularly tasks involving complex requirements, sensitive information, or highly contextual decision-making. One expert noted that the challenge is not only understanding how AI works internally but knowing when it is appropriate to rely on it. This suggests that AI tools should provide clearer signals about their scope, limitations, and appropriate contexts of use.

4.3.5 Reliability and Error Recovery. Participants also highlighted practical issues related to reliability, including repetitive outputs, hallucinations, and systems getting stuck in loops. These failures were described as frustrating because they interrupt workflow continuity and require manual correction.

5 AI-ASSISTED DESIGN FRAMEWORK

The development of the framework was grounded not only in insights from the literature review and the formative expert study, but also in a structured design exercise conducted as part of this thesis. This exercise involved designing the same feature twice—once without AI assistance and once with AI support—to observe how AI alters the cognitive and practical dynamics of early-stage UX design. The task focused on creating the user flow and initial design artefacts for a loyalty feature in a mobile application. For each condition, I created a separate workspace in the design tool, established clear rules

for what was permitted, and maintained a detailed log documenting time spent, artefacts produced, decision points, moments of friction, and, in the AI condition, the prompts used and the extent to which AI suggestions were accepted or rejected.

This comparative exercise provided first-hand insight into the subtle but significant ways in which AI reshapes design reasoning. Working without AI highlighted the manual structuring, iterative sense-making, and deliberate decision-making that typically characterise early exploratory design. In contrast, the AI-assisted condition revealed tendencies toward cognitive offloading, shifts in direction prompted by AI-generated suggestions, and challenges in articulating visual intent through textual prompts. The contrast between the two workflows made visible the points at which AI accelerates progress, the moments where it introduces ambiguity or misalignment, and the situations in which it risks distancing designers from their own reasoning.

These observations aligned closely with the themes identified in the expert interviews, particularly regarding over-automation, trust calibration, and uncertainty about appropriate use. By triangulating insights from the literature, expert perspectives, and this practice-based comparison, it became possible to identify recurring patterns in how designers work with and without AI support. These patterns directly informed the structure of the framework.

5.1 Motivation

The framework responds to a central challenge identified throughout this thesis: AI tools can accelerate early-stage work, yet their interaction models often risk diminishing human engagement, obscuring system assumptions, or encouraging premature acceptance of generated results. Rather than automating design tasks, the framework aims to support designers in using AI selectively and intentionally, ensuring that human judgment remains central to the workflow.

5.2 Foundational Themes

The framework is built on four thematic foundations that shape its structure. ***The first theme*** emphasizes the importance of keeping designers actively involved in the reasoning process. AI assistance should not replace engagement but instead scaffold it. ***The second theme*** highlights the need for visibility into system assumptions and reasoning. Designers must be able to understand why an output was produced in order to evaluate it effectively. ***The third theme*** focuses on treating AI-generated content as provisional. Outputs require critique, verification, and contextual interpretation rather than direct adoption. ***The final theme*** underscores the role of context. AI systems operate best when the boundaries of the task, the available information, and the relevant constraints are clearly defined. These themes provide the conceptual grounding for the workflow model introduced below.

5.3 Framework Structure

The framework is organized as a hybrid model that combines a sequence of workflow stages with a set of cross-cutting components. The stages describe how designers progress through an AI-assisted task, while the components establish conditions that should remain present throughout the interaction.

5.3.1 Workflow Stages.

- **Stage One: Intent.** The workflow begins with clarifying the designer's intent. Designers articulate the goal, constraints, and intended outcomes, establishing a foundation for meaningful AI

support. Although closely related to context-setting, intent focuses on *what the designer aims to achieve*.

- **Stage Two: Context.** Once the intent is defined, designers specify the contextual boundaries of the task. This includes describing the product domain, relevant user needs, existing design elements, and any constraints the AI should consider. Whereas intent expresses the direction of the task, context defines *what information the AI should rely on*.
- **Stage Three: Options.** With intent and context established, designers move into generating options. Rather than requesting a single solution, the workflow encourages the creation of multiple alternatives that can be compared, contrasted, and refined. AI contributes by proposing directions, highlighting trade-offs, or offering structural variations.
- **Stage Four: Critique.** These options are then examined through critique and verification. Designers review assumptions, identify potential issues, and evaluate the outputs against relevant criteria such as usability, feasibility, or accessibility. This stage positions AI-generated content as material for reflection rather than as definitive answers.
- **Stage Five: Improve.** After evaluation, designers select a direction and apply the chosen changes. Improvements may involve adopting AI suggestions, modifying them, or introducing new ideas. The human remains responsible for interpreting the suggestions and making the final decision, reinforcing the principle that AI supports but does not replace design judgment.
- **Stage Six: Trace.** The final stage involves documenting the trace of the interaction. Recording how AI was used, which suggestions were adopted or rejected, and the reasoning behind key decisions helps preserve accountability and supports future revisitation of the design process.

5.3.2 Cross-Cutting Components. Several components operate across all stages of the workflow. These include mechanisms for clarifying the type of assistance being requested, defining the information available to the system, and ensuring that AI outputs remain steerable and open to refinement. Additional components support the communication of assumptions and uncertainty, integrate verification into the workflow, and provide recovery paths when outputs diverge from user intent. A trace log maintains a record of interactions and decisions, supporting transparency and long-term accountability.

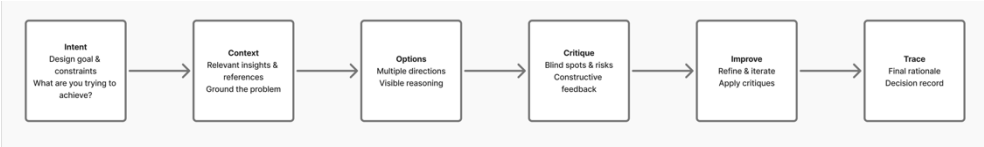


Figure 1: Diagram of the reflective design workflow, illustrating the six sequential stages (Intent, Context, Options, Critique, Improve, Trace) and the cross-cutting components that operate across the entire process. Diagram was designed by the author

6 AI DESIGN CO-PILOT ASSISTANT

To operationalise the proposed framework, I developed a functional web-based prototype designed to support reflective, early- and mid-stage design decision-making. Rather than automating design tasks, the system scaffolds the designer's reasoning process by generating alternatives, exposing blind spots, and enabling structured reflection. The prototype implements the six-stage workflow introduced in the

previous chapter and serves as the primary vehicle for examining how AI can support design work without diminishing human agency.

The workflow is operationalised in six distinct steps—Intent, Context, Options, Critique, Improve, and Trace—each supporting a different form of reasoning and interaction. Treating these as separate stages enables the system to guide designers more precisely and to generate targeted options, critiques, and improvements.

As illustrated in Figure 2, the system is composed of two main layers: the interaction flow that structures the six-stage process (Fig. 2A), and the working environment where options, critiques, and the final trace are explored and consolidated (Fig. 2B).

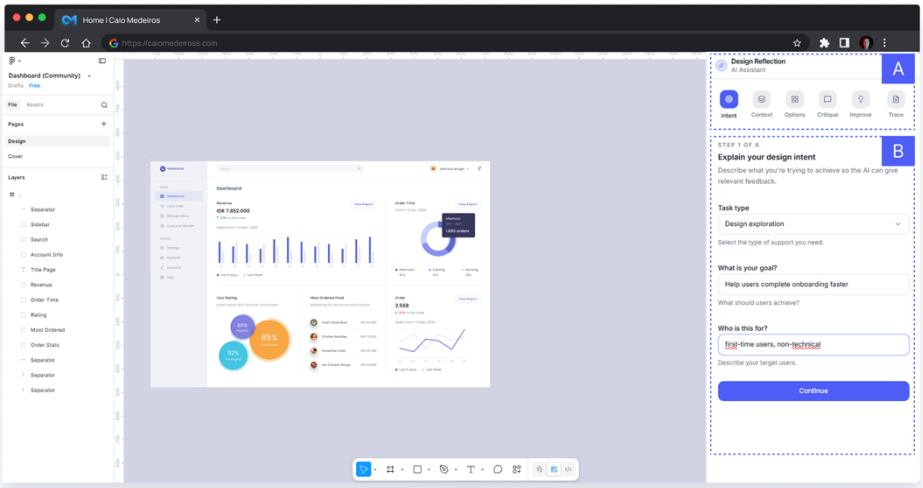


Figure 2: Overview of the prototype. (A) Interaction flow showing the six-stage navigation. (B) Working environment where options, critiques, and the final trace are explored and consolidated. Although the functional prototype used in the study was not integrated with Figma, this figure illustrates how the final product would appear within a design environment.

6.1 System Architecture

The prototype follows a client–server architecture consisting of a modular frontend and a stateless backend. The frontend orchestrates the six-stage workflow, manages global state through a custom React Context, renders options and critiques, animates the model's reasoning, and handles drag-and-drop interactions. It is implemented using React, TypeScript, Vite, TailwindCSS, and Framer Motion. The backend, implemented in Node.js and Express, receives structured context from the frontend, forwards requests to a Groq-hosted LLaMA-based model, and returns structured responses containing options, reasoning, critiques, and summaries. This separation ensures modularity and allows the backend to be replaced or extended without altering the user interface. Data flows through the system in a structured sequence: designers define intent and context, the frontend sends a JSON request to the backend, the model returns structured output, and the frontend renders options, reasoning, critiques, and the final trace.

6.2 Core Features

- ***Intent Definition:*** The first feature of the system prompts designers to articulate their goal or problem in a structured manner (Fig. 3A). This stage ensures that the AI receives a clear starting point and prevents ambiguous or overly broad prompts. By requiring designers to express intent explicitly, the system reinforces the framework's emphasis on grounding AI assistance in human-defined objectives.
- ***Context Input:*** The second feature allows designers to provide rich contextual information, including constraints, connected elements, hypotheses, user insights, and relevant background (Fig. 3B). This open-ended input supports situated, context-aware responses rather than generic suggestions. It also encourages designers to externalise assumptions, which later stages of the workflow can interrogate and refine.
- ***Option Generation:*** Once intent and context are established, the system generates three to five design options, each containing a title, summary, underlying principle, trade-off, and reasoning (Fig. 3C). The reasoning is displayed through a progressive animation inspired by contemporary conversational AI interfaces, making the chain-of-thought visible without overwhelming the user. This design choice supports transparency and aligns with the framework's emphasis on making system reasoning accessible and interpretable.

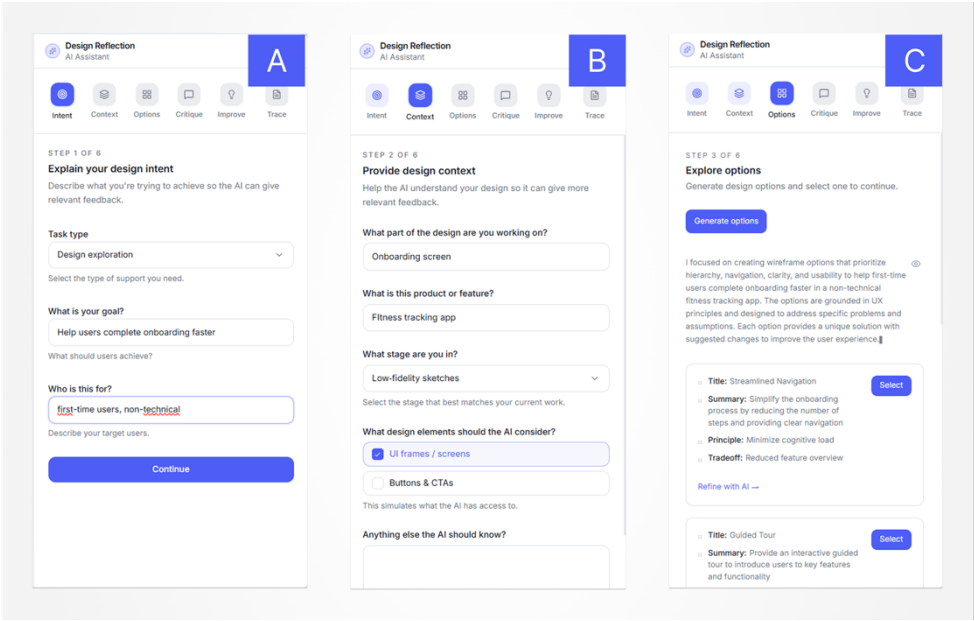


Figure 3: Overview of the first three core features. (A) Intent definition interface. (B) Context input interface. (C) Option generation interface.

- ***Mix & Match:*** To reflect real design practice, the prototype includes a Mix & Match mechanism that allows designers to combine elements from different options (Fig. 4A). Titles, summaries, principles, and trade-offs can be rearranged through a drag-and-drop interface, enabling designers to construct hybrid solutions rather than accepting any option wholesale. This feature

operationalises the framework's focus on maintaining human agency and supporting exploratory recombination.

- ***Critiques:*** The system provides critiques for each option, addressing usability, accessibility, consistency, risks, and potential blind spots (Fig. 4B). These critiques encourage reflective evaluation rather than passive acceptance of AI-generated suggestions. Designers can request critiques at any point, allowing them to revisit earlier ideas with additional perspective and reinforcing the framework's emphasis on verification.
- ***Trace:*** The final feature is the trace, an editable summary where designers consolidate their decision and reasoning (Fig.4C). The trace captures which elements were selected, how they were combined, and why certain choices were made. This record supports transparency, accountability, and knowledge transfer, and aligns with the framework's emphasis on documenting the evolution of reasoning.

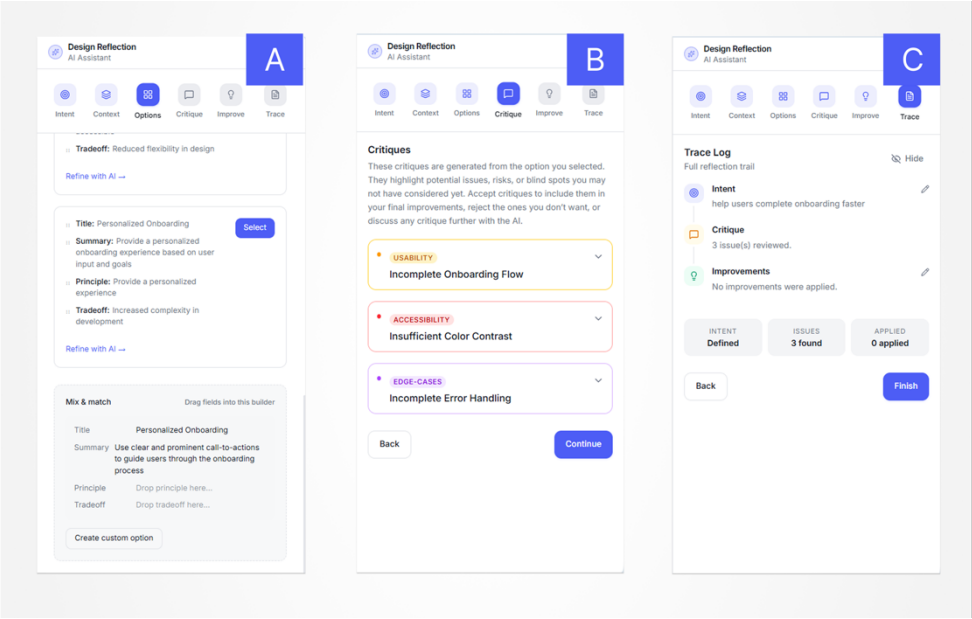


Figure 4: Overview of the final three core features. (A) Mix & Match interface for recombining elements across options. (B) Critique interface showing usability, accessibility, and risk-focused evaluations. (C) Trace interface where designers consolidate selected elements and articulate their final reasoning.

7 EVALUATION

This chapter presents the evaluation of the AI Design Co-Pilot prototype through a two-part user study consisting of a low-fidelity testing session followed by a high-fidelity testing session. Both sessions examined the same underlying six-stage workflow—Intent, Context, Options, Critique, Improve, and Trace—while differing in prototype maturity and interaction fidelity. The overarching aim was to assess whether the workflow supports reflective design, preserves designer agency, and enables meaningful human–AI collaboration. In both sessions, the prototype was not integrated with Figma; participants

worked with speculative scenarios, and the emphasis remained on interaction flow, prompting, and LLM-generated feedback rather than visual fidelity.

7.1 Low-fidelity testing session

7.1.1 Participants. Four participants took part in the low-fidelity session. They represented a mix of UX designers, non-designers, and students with varying levels of familiarity with design tools and AI systems. All participants had interacted with generative AI tools before, though none had used the prototype. This diversity allowed the study to capture differences in how design expertise shapes interpretation of the workflow.

7.1.2 Study procedure. The low-fidelity study evaluated an early clickable prototype that represented the full six-stage workflow—Intent, Context, Options, Critique, Improve, and Trace—although in a simplified form. Each session lasted approximately 10–15 minutes and followed a think-aloud protocol. Before beginning, participants completed a short pre-test questionnaire covering their familiarity with UX/UI design, experience with tools such as Figma, and general interaction with digital interfaces.

Participants were then guided through each of the six stages. In the Intent and Context stages, they were asked to articulate a design goal and provide the contextual information the AI should consider. In the Options stage, they reviewed a small set of AI-generated directions. The Critique stage required them to examine AI-generated critique cards, and in the Improve stage they responded to these critiques by selecting or writing an improvement. Finally, they reviewed a Trace summarising their intent, context, options, critiques, and improvements. The session concluded with a short interview probing workflow clarity, confusion points, and perceived usefulness.

7.1.3 Testing results. The low-fidelity session revealed that participants understood the overall logic of the six-stage workflow and described the interface as clean and trustworthy. However, several aspects of the system created confusion. The product context was unclear, and labels such as “Select elements for AI” caused uncertainty about what the system expected from them. Participants struggled to understand how their intent and context inputs shaped the AI’s subsequent reasoning. One participant noted, “I get the idea, but I’m not always sure what the system expects me to write here,” indicating that the mental model of the tool was not yet fully established.

The step structure felt logical but overly rigid. Participants wanted to skip steps, merge similar steps, or revisit earlier ones. Intent and Context were perceived as too similar, and the Trace stage felt more appropriate for later design stages rather than early ideation. Although critiques were generally seen as clear and constructive, participants expressed a desire for more conversational iteration. As one participant explained, “I want to ask the AI to elaborate, not just read static cards.”

Participants also emphasised the importance of real-time visual feedback. The lack of Figma integration made it difficult to imagine the impact of applying improvements. The Trace was appreciated for its transparency, but participants wanted it to be optional, editable, and better aligned with their workflow.

Overall, the low-fidelity session confirmed the value of the six-stage workflow but highlighted the need for clearer communication, more flexible navigation, and more interactive critique and improvement flows.

7.2 Iterations after low-fidelity session

Insights from the low-fidelity session informed several key design changes. Labels were rewritten for clarity, and one-line descriptions and placeholder examples were added to each step to help users understand what type of input was expected. Tooltips were introduced to explain key terms, and the relationship between early stages was clarified. To address the desire for flexibility, non-linear navigation was enabled through skip buttons and a sidebar. An introductory section was added to explain what the tool is, what it does, and how AI uses input. A refinement chatbot was introduced to support conversational iteration, and the trace log was made editable. These changes were incorporated into the high-fidelity prototype tested in the second session.

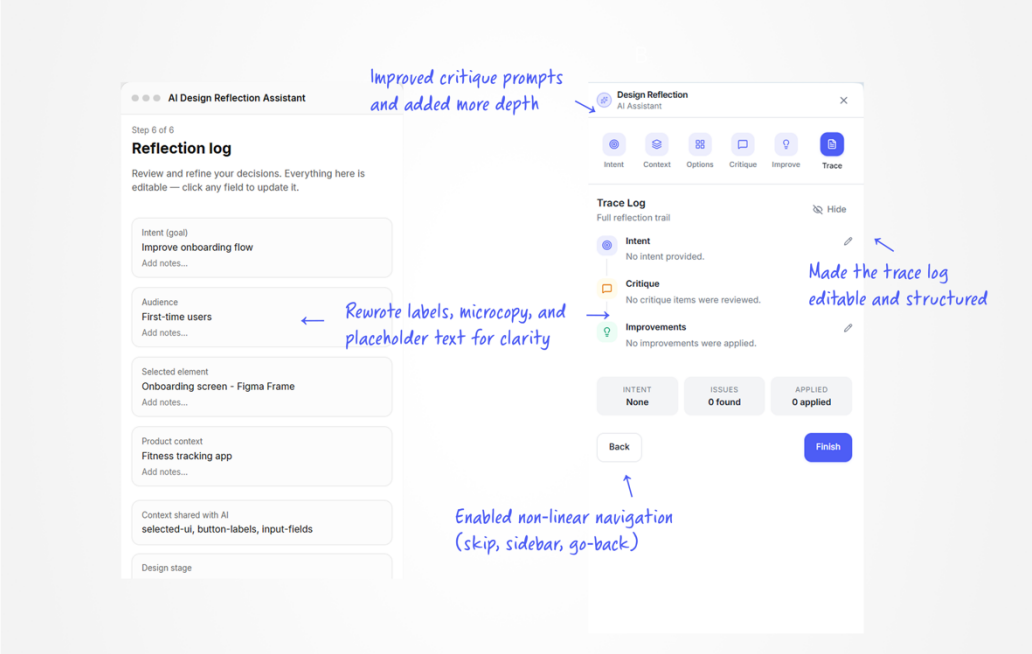


Figure 5: Comparison between the initial low-fidelity prototype and the improved iteration incorporating insights from the first evaluation session.

7.3 High-fidelity testing session

7.3.1 Participants. Eight participants took part in the high-fidelity session, including UX designers, product designers, researchers, and students. Their experience ranged from one to nine years, and their familiarity with AI tools varied from low to high. All participants had used generative AI tools before but none had interacted with the prototype.

7.3.2 Study procedure. The high-fidelity study evaluated the full six-stage workflow using a coded prototype. Participants completed a realistic design task—exploring two to three onboarding directions for a fintech app—while thinking aloud. The study included a pre-test questionnaire, five task scenarios (design exploration, evaluation, user research, UX writing, and product strategy), a detailed post-test questionnaire covering eight UX dimensions, and a semi-structured interview probing agency, collaboration, clarity, trust, and improvement suggestions.

Participants were encouraged to comment on mismatches between expectations and output, the clarity of each step, and the usefulness of critiques, improvements, and reasoning. They were also asked to reflect on whether the AI supported or constrained their thinking.

7.3.3 Testing results. Participants consistently praised the structured workflow, sense of control, and collaborative feel of the AI. Many described the AI as a supportive partner rather than a decision-maker. One participant noted, "It feels like the AI is giving me angles to think about, but I still decide what matters," while another emphasised, "I could always add my own inputs and wasn't forced to choose an option."

Despite these strengths, several cross-participant pain points emerged. Critiques were often perceived as too generic, with participants describing them as "reminders rather than deep insights." Improvements were sometimes unclear or buggy, and participants wanted more variety in the options generated. The AI occasionally reintroduced rejected ideas, which undermined the sense of agency. As one participant stated, "If I say no, I want the system to drop it."

Participants also expressed a strong desire for visual integration. Many expected wireframes, mock-ups, or the ability to upload their own designs. Transparency was another recurring theme: participants wanted to understand why the AI made certain suggestions and how options differed. One participant remarked, "I want to see how the AI thinks," highlighting the need for visible reasoning.

Navigation issues also surfaced. Participants requested a global "Go back" button, smoother transitions from the trace log, and more flexible movement between stages. These issues underscored the need for a workflow that accommodates different working styles.

Overall, the high-fidelity session validated the workflow but revealed the need for deeper reasoning, more flexible navigation, richer options, and visual integration.

7.4 Iterations after high-fidelity session

The high-fidelity session informed several further improvements. The distinction between critique and improvement was clarified, and "Apply," "Suggest," and "Reject" controls were added to strengthen agency. The system was updated to respect rejected suggestions and avoid reintroducing them. Critique variety was expanded and grounded in design stage, and Mix & Match functionality was introduced to allow recombination of elements across options. Visible AI reasoning was added to explain why suggestions might help, and a global "Go back" button improved navigation. Support for uploading files and attaching context was prepared, and improvement-related bugs were fixed. These iterations responded to insights from the formative study, aiming to enhance agency, transparency, and adaptability in ways that better reflect real-world design workflows.

8 DISCUSSION

The findings from both studies suggest that the value of AI in design may lie less in automating tasks or producing polished outputs, and more in how it can scaffold designers' reasoning, surface blind spots, and support more deliberate decision-making. Participants often described the workflow as functioning more like a thinking partner than a production tool, indicating that reflective AI assistance may address needs not fully met by existing generative systems.

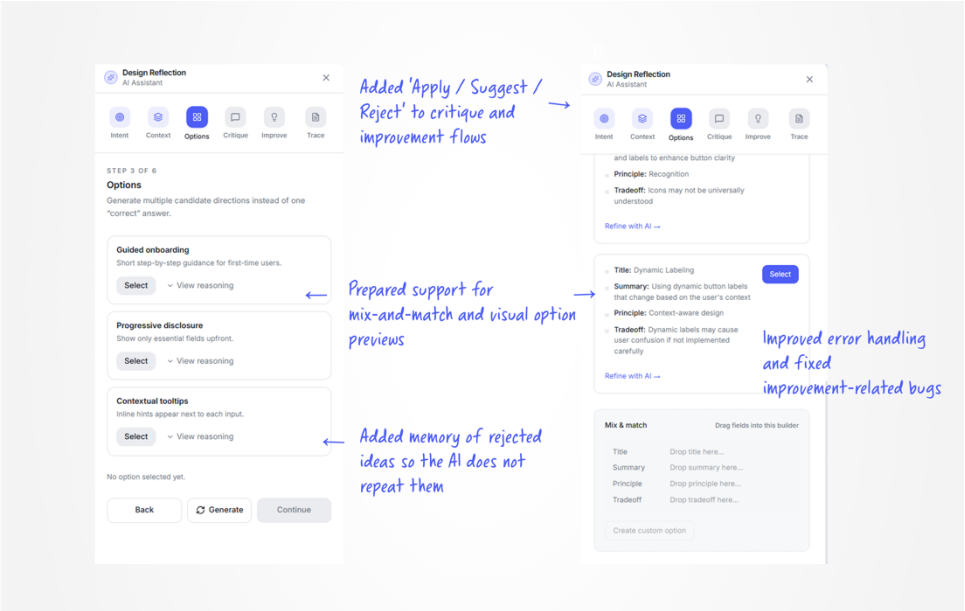


Figure 6: Comparison between the initial high-fidelity prototype and the improved iteration incorporating insights from the second evaluation session. Diagram designed by the author.

8.1 Interpretation of the Findings

The findings indicate that designers do not primarily seek AI systems that generate solutions on their behalf; instead, they value tools that help them think. Participants repeatedly described the AI as offering "angles to think about" or "perspectives I hadn't considered," emphasising that the system's usefulness came from structuring reflection rather than producing answers. This aligns with the central premise of the framework: the question is not "How good is AI at brainstorming or wireframing?" but "How should designers think with AI?"

The six-stage workflow proved effective in supporting this reflective stance. Intent and Context helped participants articulate what mattered before generating options, while the Critique and Improve stages encouraged them to interrogate assumptions and justify decisions. The Trace stage, although more relevant in later design phases, reinforced accountability and transparency. Importantly, participants felt in control throughout the process, even when the AI offered strong suggestions. This suggests that structured workflows can mitigate the risk of over-reliance or automation bias by making human judgment explicit at each step.

At the same time, the findings reveal tensions. Participants wanted deeper, more context-sensitive critiques and clearer reasoning behind suggestions. They also expected the system to adapt to different design stages and to respect rejections without reintroducing discarded ideas. These expectations highlight the limits of current LLM behaviour and the need for more robust mechanisms for grounding, memory, and stage-awareness.

8.2 Relation to Prior Work

The findings of this thesis resonate strongly with existing research on the limitations of generative AI in design. Prior work shows that most AI design tools focus on producing interface-level outputs such as

wireframes, layouts, or stylistic variations, rather than supporting upstream activities such as problem framing, contextual understanding, or design reasoning [12, 21]. Studies by Stahl et al. and Lee et al. highlight that generative tools accelerate early exploration but offer limited support for deeper cognitive processes, a limitation also observed in controlled experiments showing that large language models do not reliably improve problem reframing or contextual reasoning [20]. The results of this thesis reinforce this gap: participants consistently sought help not with generating screens, but with clarifying intent, evaluating trade-offs, and understanding the implications of their decisions.

Similarly, research on human–AI collaboration emphasises the importance of transparency, predictability, and user control as prerequisites for trust [5]. Designers often misjudge AI capabilities, leading to miscalibrated trust or over-reliance, a challenge also documented in studies of collaborative creativity where designers either overestimate or underestimate AI competence [2]. The structured workflow developed in this thesis directly addresses these concerns by making the designer's intent explicit, separating context from output generation, and requiring human interpretation at each stage. Participants' strong sense of agency—expressed in comments such as "I still decide what matters"—suggests that the workflow successfully mitigates the risks identified in prior work.

The findings also speak to the limitations of prompt-based interaction. Prior studies show that designers find text prompting misaligned with visual and iterative design practices, and that translating visual intent into precise textual descriptions can be difficult [21]. Research on accessibility further demonstrates that increasing prompt complexity does not reliably improve accessibility outcomes, even when requirements are explicitly stated [10, 13]. The prototype developed in this thesis offers an alternative interaction model: instead of relying on a single prompt, it structures the interaction into multiple stages that guide the AI's reasoning and reduce the cognitive burden on the designer. This contributes a concrete example of how AI-assisted design tools can move beyond prompt engineering toward more structured, reflective workflows.

8.3 Implications for Design Practice

The results suggest that reflective AI tools can strengthen early- and mid-stage decision-making by slowing designers down in environments that often prioritise speed and output. Participants described the workflow as helping them "think better," "structure ideas," and "see the problem from different angles." This indicates that AI can play a meaningful role in supporting the cognitive work of design, not just the generative or operational aspects.

However, this reflective orientation introduces a tension with industry incentives. Many commercial AI design tools emphasise automation, efficiency, and rapid output generation. A system that encourages deliberation, critique, and traceability may therefore be at odds with dominant narratives of speed and productivity.

The desire for visual integration also points to a broader implication: reflective AI tools must coexist with designers' existing ecosystems. Participants expected the ability to upload wireframes, manipulate a canvas, or preview changes visually. This suggests that reflective AI should not replace existing tools but augment them, embedding reasoning support within familiar workflows.

Overall, the results underscore the importance of maintaining human agency in AI-assisted design. Rather than replacing designers or accelerating production, reflective AI systems can help designers articulate intent, examine assumptions, and make more deliberate decisions. This positions AI not as a generator of solutions, but as a collaborator in the reasoning process.

8.4 Ethical, Sustainability, and Societal Considerations

Sustainability is not a central focus of this thesis, as the work concerns AI-assisted design workflows rather than physical products or resource-intensive systems. Nevertheless, the use of generative AI introduces relevant sustainability considerations, particularly regarding the computational resources required by AI systems. These considerations are outside the scope of the present work and are therefore not evaluated directly. Instead, the thesis focuses primarily on the social and ethical implications of integrating AI into design workflows.

A key concern is the potential impact of AI-assisted tools on designers' agency, skills, and long-term working practices. As discussed in the previous chapters, excessive automation may reduce opportunities for designers to reason through problems themselves and may encourage users to accept AI-generated solutions without sufficient reflection. This raises broader societal concerns about skill development, professional responsibility, and the changing role of designers as AI becomes increasingly integrated into creative work.

Ethical considerations are therefore incorporated into the proposed framework through principles such as maintaining human control, making assumptions and uncertainty visible, protecting sensitive context, and requiring verification before applying AI-generated suggestions. The framework also emphasizes that AI should support rather than replace human decision-making. These principles aim to reduce the risk of over-reliance on AI while preserving designers' responsibility and ownership of their work.

Overall, while sustainability is not the primary focus of the thesis, the work recognizes that AI-assisted design has social, ethical, and ecological dimensions. The framework primarily addresses the social and ethical dimensions at the interaction level, while the broader environmental impact of AI remains an area for further investigation.

8.5 Use of Generative AI in This Thesis

Generative AI tools were used in this thesis to support writing-related tasks such as grammar checking, language polishing, and improving clarity of expression. All theoretical framing, methodological decisions, design concepts, analyses, and interpretations presented in this thesis are entirely my own.

9 LIMITATIONS AND FUTURE WORK

This project has several limitations that shape the scope of its contributions and point toward opportunities for future research. The interview study involved a small and relatively homogeneous group of participants, which limits the breadth of perspectives represented.

The prototype developed for this thesis represents only one interpretation of the proposed framework. Time constraints limited the exploration of alternative interaction flows, visual designs, and system behaviours, meaning that the evaluation reflects the performance of a single implementation rather than the broader design space implied by the framework. Additional prototypes could reveal different strengths, weaknesses, or emergent patterns of use, particularly in how designers engage with verification, context-setting, and iterative refinement.

A further limitation of the current prototype is its lack of integration with existing design and productivity platforms. The system operates as a standalone web application, meaning it cannot access real design artefacts, ongoing project files, or the contextual information embedded in tools that designers already use. As a result, the evaluation does not reflect how the framework would perform when embedded within established workflows such as Figma, VS Code, Notion, or other environments where designers

make decisions, document rationale, and iterate on artefacts. Integrating the framework as a plugin or extension would significantly increase ecological validity by allowing AI assistance to operate directly on live design materials, codebases, or documentation

Another limitation concerns the short-term nature of the evaluation. The study focused on immediate impressions and short interactions, which does not capture how designers might adapt their workflows over time, how trust and reliance evolve with repeated use, or how the framework performs under real project constraints. Longitudinal studies would be valuable for understanding how AI assistance becomes integrated into everyday routines, how verification strategies develop, and how designers negotiate the balance between automation and control over extended periods.

Future work could also expand the prototype's technical capabilities. Participants expressed interest in more adaptive behaviours, such as AI that responds to user edits, provides concise summaries of pain points, or offers structured recommendations at different stages of the workflow. Integrating the system with existing design tools such as Figma or Canva would increase ecological validity and allow designers to use AI assistance within familiar environments. Additional features—such as support for file attachments, screenshots, or reusable templates—could further enhance practical value and reveal new opportunities for supporting reflective design practice.

Finally, the framework currently focuses on individual workflows, but many design processes involve collaboration across teams, disciplines, and time zones. Extending the framework to support multi-designer or asynchronous collaboration would require examining how AI-generated content is shared, interpreted, and revised across distributed settings.

10 CONCLUSION

This thesis set out with an exploratory ambition: to understand how designers might useAI in a more meaningful and responsible way within the design process. At the beginning of the project, the goal was intentionally open, a broad question of how AI could support design work without replacing human thinking. Through extensive exploration, interviews, and iterative reflection, the research questions became clearer: the challenge is not how good AI is at individual design tasks, but how designers should engage with AI as a whole.

The results show several positive effects. The proposed framework and prototype demonstrate that AI can support designers by offering structured reflection, alternative perspectives, and critique without taking control of the process. Designers remain the primary decision-makers, and AI acts as a thinking partner rather than an automated generator. This reinforces the insight thathuman reasoning must always come first, and that AI should amplify, not replace designer agency.

However, the work also revealed drawbacks. Current AI systems are optimised for automation and efficiency, which makes reflective, human-first tools less aligned with mainstream industry incentives. While the concept is valuable for design ethics and long-term thinking, it may face adoption challenges in corporate environments that prioritise speed and output. I hope this thesis contributes to ongoing efforts to develop AI-supported design environments where human insight, contextual reasoning, and machine assistance evolve together in a balanced and accountable way.

ACKNOWLEDGMENTS

I would first like to express my deepest gratitude to my family. To my parents, Sebastián Lendínez Ruíz and Carmen Ibáñez Polo, and to my brother, Adrián Lendínez Ibáñez: thank you for raising me, loving

me, and supporting every decision I have ever made. Thank you for shaping the person I am today, for believing in me even when I doubted myself, and for being the very best part of my life. To my grandmother, Matilde Ruiz, whose stories, strength, and imagination have inspired me since I was a child. Thank you for encouraging me to live fully, to seek adventure, and to make these years in Sweden some of the most meaningful of my life. I am sincerely grateful to my supervisor, Ran Zhou, for listening, guiding, and challenging me with thoughtful ideas. Your support and insight have made this thesis the best it could be. To my friends scattered across the world—those who have accompanied me since childhood and those I met along the way in every stage of my life— thank you. Thank you for making my youth unforgettable, for bringing joy into the moments when I was overworking, for listening to me, for sharing adventures, and for becoming my chosen family while living abroad. To my friends in Spain, Portugal, France, Italy, Austria, Germany, Sweden, China, Vietnam, and Korea: you have shaped my journey more than you know. Thank you all for walking with me through this chapter. This thesis is as much yours as it is mine.

REFERENCES

[1] Cansu Akverdi and Gökçe Elif Baykal. 2024. Generative AI Tools in Design Fields: Opportunities and Challenges in the Ideation Process. In *Adjunct Proceedings of the Nordic Conference on Human-Computer Interaction*. ACM. https://doi.org/10.1145/3677045.3685445

[2] Mia Magdalena Bangerl, Leonie Disch, Tamara David, and Viktoria Pammer-Schindler. 2025. CreAItive Collaboration? Users' Misjudgment of AI-Creativity Affects Their Collaborative Performance. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3706598.3713886

[3] Rishi Bommasani, Drew A. Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S. Bernstein, Jeannette Bohg, Antoine Bosselut, Emma Brunskill, et al. 2021. On the Opportunities and Risks of Foundation Models. *arXiv preprint arXiv:2108.07258* (2021).

[4] Daniel Buschek, Michael Zürn, and Florian Alt. 2021. The Impact of AI Autocompletion on User Agency. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3411764.3445311

[5] Beenish Moalla Chaudhry. 2024. Concerns and Challenges of AI Tools in the UI/UX Design Process: A Cross-Sectional Survey. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3613905.3650878

[6] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, et al. 2021. Evaluating Large Language Models Trained on Code. In *arXiv preprint arXiv:2107.03374*.

[7] Xiang 'Anthony' Chen, Tiffany Knearem, and Yang Li. 2025. The GenUI Study: Exploring the Design of Generative UI Tools to Support UX Practitioners and Beyond. In *Proceedings of the ACM Designing Interactive Systems Conference*. ACM. https://doi.org/10.1145/3715336.3735780

[8] Graham Dove, Kim Halskov, Jodi Forlizzi, and John Zimmerman. 2017. UX Design Innovation: Agency and Responsibility in AI Systems. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3025453.3025739

[9] Nishit Gajjar, Vinoth Pandian Sermuga Pandian, Sarah Suleri, and Matthias Jarke. 2021. Akin: Generating UI Wireframes From UI Design Patterns Using Deep Learning. In *Proceedings of the International Conference on Intelligent User Interfaces*. ACM. https://doi.org/10.1145/3397482.3450727

[10] Alexandra-Elena Gurita. 2025. Understanding and Improving Accessibility in AI-Generated Interfaces through Interactive Prompt Engineering Methods. In *Companion Proceedings of the International Conference on Intelligent User Interfaces*. ACM. https://doi.org/10.1145/3708557.3716347

[11] Abidullah Khan, Atefeh Shokrizadeh, and Jinghui Cheng. 2025. Beyond Automation: How Designers Perceive AI as a Creative Partner in the Divergent Thinking Stages of UI/UX Design. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3706598.3713500

[12] Kate Sangwon Lee, Mijin Choi, and Eda Yasmin Asni. 2025. AI Opportunity Cards: Developing a Toolkit for AI as a Design Material. In *Proceedings of the International Conference on Information Technology for Social Good*. ACM. https://doi.org/10.1145/3748699.3749817

[13] Katelyn Leedy, Makenzie Preston, Jinjuan Feng, and Jeba Rezwana. 2025. Accessibility in the Age of Generative AI Web Based Builders: Evaluating Web Design Tools for Inclusive Practices. In *Proceedings of the International Academic Mindtrek Conference*. ACM. https://doi.org/10.1145/3757980.3757989

[14] Vivian Liu and Lydia B. Chilton. 2023. Designing with Generative Models: Supporting Creative Exploration. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3544548.3580701

[15] Ryan Louie, Justin Li, and Walter S. Lasecki. 2020. Novice and Expert Perspectives on AI-Assisted Creative Work. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3313831.3376810

[16] Pavel Okopnyi, Oda Elise Nordberg, and Frode Guribye. 2024. Against Generative UI. In *Proceedings of the Halfway to the Future Symposium*. ACM. https://doi.org/10.1145/3686169.3686184

[17] OpenAI. 2023. *GPT-4 Technical Report*. Technical Report. OpenAI. https://arxiv.org/abs/2303.08774

[18] Savvas Petridis, Michael Terry, and Carrie J. Cai. 2024. PromptInfuser: How Tightly Coupling AI and UI Design Impacts Designers' Workflows. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3643834.3661613

[19] Daniel Saunders, William Odom, and Reuben Kirkham. 2023. Co-Reflection with AI: Supporting Reflective Design Practices. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3544548.3580702

[20] Joongi Shin, Anna Polyanskaya, Andrés Lucero, and Antti Oulasvirta. 2025. No Evidence for LLMs Being Useful in Problem Reframing. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3706598.3713273

[21] Anna Ståhl, Kristina Höök, Sara Eriksson, and Wonjun Lee. 2023. Design Principles for AI UX Delivering User Value for Samsung Galaxy. In *Proceedings of the 25th International Conference on Mobile Human-Computer Interaction*. ACM. https://doi.org/10.1145/3565066.3608243

[22] Priyan Vaithilingam, Tianyi Zhang, and Elena L. Glassman. 2022. Expectation vs. Experience: Evaluating Code Generation Tools with Programmers. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3491102.3502030

[23] Zhutian Wu, Yang Li, Jiajun Zhu, Yuhang Li, and Yang Li. 2023. AI Ideation: Designing with Generative Models in Early-Stage Creative Work. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3544548.3580700

[24] Qian Yang, Ece Kamar, F. Maxwell Harper, and Eric Horvitz. 2018. Investigating Human-AI Interaction: A Case Study of Writing Assistance. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3173574.3174223

[25] Qian Yang, Aaron Steinfeld, and John Zimmerman. 2020. Re-examining Human-AI Interaction: Designing for Complementarity. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3313831.3376817

[26] Weitao You, Yinyu Lu, Zirui Ma, Nan Li, Mingxu Zhou, Xue Zhao, Pei Chen, and Lingyun Sun. 2025. DesignManager: An Agent-Powered Copilot for Designers to Integrate AI Design Tools into Creative Workflows. *ACM Transactions on Graphics* (2025). https://doi.org/10.1145/3730919

TRITA – EECS-EX 2024:0000
Stockholm, Sweden 2026

www.kth.se

€€€€ FOR DIVA €€€€

{
”Author1”: { ”Last name”: ”Lendínez Ibáñez”,
”First name”: ”Marta Casandra”,
”Local User Id”: ”u1jz6z75”,
”E-mail”: ”mcli2@kth.se”,
”organisation”: {”L1”: ”School of Electrical Engineering and Computer Science”,
}
},
”Cycle”: ”2”,
”Course code”: ”DA232X”,
”Credits”: ”30.0”,
”Degree1”: {”Educational program”: ”Master’s Programme, Interactive Media Technology, 120 credits
”
,”programcode”: ”TIMTM”
,”Degree”: ”Both”
,”subjectArea”: ”Media Technology”
},
”Title”: {
”Main title”: ”Human-Centered AI Assistance in Design Tools”,
”Subtitle”: ”Designing a Task-Aware Co-Pilot to Support UX Designers’ Exploration and Decision-Making”,
”Language”: ”eng” },
”Alternative title”: {
”Main title”: ”Människocentrerad AI-assistans i designverktyg”,
”Subtitle”: ”Att designa en uppgiftsmedveten AI-co-pilot för att stödja UX-designers utforskning och beslutsfattande”,
”Language”: ”swe”
},
”Supervisor1”: { ”Last name”: ”Zhou”,
”First name”: ”Ran”,
”Local User Id”: ”ranzhou@kth.se”,
”E-mail”: ”sa@kth.se”,
”organisation”: {”L1”: ”School of Electrical Engineering and Computer Science”,
”L2”: ”Division of Media Technology and Interaction Design” }
},
”Examiner1”: { ”Last name”: ”Laaksolahti”,
”First name”: ”Jarmo”,
”Local User Id”: ”u1civhar”,
”E-mail”: ”jarmola@kth.se”,
”organisation”: {”L1”: ”School of Electrical Engineering and Computer Science”,
”L2”: ”Division of Media Technology and Interaction Design” }
},
”National Subject Categories”: ”10201, 10209”,
”Other information”: {”Year”: ”2026”, ”Number of pages”: ”21”},
”Series”: { ”Title of series”: ”TRITA – EECS-EX” , ”No. in series”: ”2024:0000” },
”Opponents”: { ”Name”: ”A. B. Normal & A. X. E. Normalè”},
”Presentation”: { ”Date”: ”2022-03-15 13:00”
,”Language”:”eng”
,”Room”: ”via Zoom https://kth-se.zoom.us/j/ddddddddddd”
,”Address”: ”Isafjordsgatan 22 (Kistagången 16)”
,”City”: ”Stockholm” },
”Number of lang instances”: ”2”,
”Abstract[eng]”: €€€€
1abstracts €€€€,
”Keywords[eng]”: €€€€
Human-Centered Artificial Intelligence, UX Design Tools, AI-Assisted Design, Large Language Models, Human–AI Collaboration
€€€€,
”Abstract[swe]”: €€€€
2abstracts €€€€,
”Keywords[swe]”: €€€€
Människocentrerad artificiell intelligens, UX-designverktyg, AI-assisterad design, Stora språkmodeller, Människa–AI-samarbete
€€€€,
}